

SANKUR KUNDU

Software Engineer · Machine Learning Engineer

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Engineer who ships systems end to end. Co-founder and Head of Engineering at Gigzen: built both products, the Postgres they share, and the [public record of every test run](#). Shipped on npm, as a Windows app and as a signed Android build.

EXPERIENCE

Gigzen Private Limited · Co-founder & Head of Engineering

Bhubaneswar | Apr 2026 – Present

- Co-founded an MCA-registered company; lead engineering, schema to signed release. Both products share a Postgres database with **row-level security on every table**, verified by attack: a driver reads 0 of another's messages; writes get 403.
- **Populace**: built a multi-user testing platform that drives an app through its own API as simulated, authenticated users. Its first run found **5 defects in 3.5 minutes**, 3 of them in an Android app that had passed full manual testing.
- App-agnostic engine behind a 13-method adapter contract, proven on Gitea, with per-endpoint p50/p95 latency and three guards against running on production; **121 self-tests**, zero runtime dependencies. Ships on [npm](#) and as an Electron app.
- Load-tested with **200 concurrent simulated drivers: 932,455 API calls, zero failures, p95 under 900 ms**; across 3,174,576 calls in one day, measured scalability from 178 to 272 calls/sec at 100 to 300 drivers.
- **Waggle**: React, TypeScript, Capacitor on Supabase with a realtime feed, group chat, voice notes and GPS tracking map-matched to roads; an Android foreground service records with the screen off. **43 languages**, 70 currencies, 92 countries.
- Built offline-first: posts and messages queue on the device without signal and send on reconnect; 8.2 MB build.
- Debugged a recursive row-level-security policy that silently broke every chat read and a distance algorithm that paid parked drivers for GPS drift. Moved 479 KB Postgres photo blobs (9.4 MB a feed) to object storage with ~20 KB thumbnails.

Freelance Data Scientist · Contract via Upwork

Remote, client in Spain | Jun – Jul 2026

- Merged two Spanish hotel censuses (5,909 and 8,058 records; 118 and 281 columns) on national statistics identifiers in Python and Pandas, over Excel and STATA .dta sources.
- Built a Google Trends panel of 10,038 city-months: deseasonalised, z-scored and flagged for positive shocks.
- Collaborated cross-functionally with researchers and statisticians on validation and reproducibility checks.

SELECTED ML PROJECTS

- **ML Serving Platform** — A/B routing, circuit breaking and drift detection around inference; Triton, MLflow, Kubernetes.
- **Edge Model Compression** — distillation, pruning, int8 ONNX; targeted 8× smaller, measured -6.3%, published why.
- **IoT Predictive Maintenance on GCP** — LSTM failure prediction on a leak-free chronological split; BigQuery, Vertex AI.
- **DETEK, Data-Leak Detection on AWS** — BERT content classifier paired with an LSTM behavioural model; Terraform.

SKILLS

Languages: Python, TypeScript, Java, SQL, R, STATA

ML & AI: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy; deep learning (LSTM, BERT, DDPM), training and evaluation pipelines, statistical modeling, distillation, pruning, int8 quantization, ONNX, anomaly detection, LLM-assisted diagnosis

Systems & Cloud: System design, distributed systems, CI/CD, GCP (BigQuery, Vertex AI, Dataflow, Pub/Sub), AWS (Lambda, S3), FastAPI, Flask, Node.js, React, Electron, Postgres, Redis, Docker, Kubernetes, Terraform

EDUCATION & CERTIFICATIONS

Institute of Engineering and Management, Kolkata · B.Tech, Computer Science & Engineering

2022 – 2026

Certifications: Meta Data Analyst; Google Data Analytics; Google Cybersecurity; IBM Cybersecurity Essentials; Google Cloud and AWS engineering paths; Microsoft Azure AI Cloud Apps